
Unit #1

A (person's) z-score

Sam's flourishing score was 1.25 standard deviations above the mean. If flourishing is normally distributed, Sam's score is between the 84th and the 98th percentile.

Mean and standard deviation for a variable

On a Flourishing Scale that ranged from 8 to 56, the average score was 43.22. On average, people deviated from this score by 9.41 units.

Unit #2

Reliability of a scale

The Flourishing Scale had good reliability (Cronbach's Alpha = .83). These results suggest that 17% of the variance was measurement error and that on average, a participant's observed score is likely to deviate from her or his true score by 3.88 units.

Unit #3

Paired samples analysis

There was a significant difference between scores at Time 1 and Time 2 ($d = .78$, $t(52) = 3.33$, $p = .002$), with the people reporting higher rudeness before the program ($M = 18.13$, $SD = 6.75$) than after the program ($M = 12.98$, $SD = 6.43$). This represents a medium effect.

Independent samples summary

There was a significant difference between people in the exercise group and people in the control condition ($d = .81$, $t(103) = 4.12$, $p < .001$), with the people in the exercise group reporting more emotional wellbeing ($M = 26.38$, $SD = 4.93$) than people in the control condition ($M = 22.47$, $SD = 4.77$). This represents a large effect.

Unit #5

Regression results

In regression analysis, exercise was predicted using income, addressing barriers, and goal clarity. The three predictors explained a significant portion of variance in exercise ($R^2 = .11$, $F(3, 478) = 28.22$, $p < .001$). Correlations and standardized betas are reported in Table 1. There was a medium positive correlation between goal clarity and exercise, and goal clarity explained unique variance in exercise after controlling for the effects of the other variables in the equation. Although there was a small positive correlation between addressing barriers and exercise, the effect became non-significant after controlling for other variables. Notably, income was not significantly related to exercise.

Table 1

Predicting exercise

	Correlation	Standardized beta
Goal clarity	.39***	.36***
Addressing barriers	.24***	.05
Income	.08	.02

*** $p < .001$